Report on facility location recommendation for the Boeing Company

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Abstract

This report builds upon the Project 1on financial analysis of Boeing Company and presents a detailed evaluation and recommendation for the location of a new U.S.-based production facility. After assessing multiple site options, Charleston, South Carolina and Wichita, Kansas were selected as the final recommendations based on strategic, operational, and financial considerations. Financial modeling using a 30-year CFAT-based forecast, IRR, NPV, Payback, and Annual Worth metrics were conducted. Wichita emerges as the optimal location, offering superior financial returns and strategic benefits.

Company Overview

The Boeing Company, an iconic name in global aerospace and defense manufacturing, has faced significant financial and operational headwinds in recent years. My Project 1 analysis detailed how Boeing's financial performance has been impaired due to a series of supply chain constraints, pandemic-driven downturns, legal setbacks from the 737 MAX incidents, and ongoing pension liabilities. Key financial ratios such as ROI and ROA have remained negative since 2019, reflecting a broader decline in operational efficiency. Liquidity concerns were highlighted by a low quick ratio, suggesting Boeing's need for a strategic recalibration.

Despite these challenges, Boeing retains a strong brand presence, long-term government contracts, and a solid base of technological expertise. These assets position Boeing to recover effectively if it takes steps to realign its operations. As part of that recovery, this project recommends a new U.S.-based facility that would enhance production capacity, improve logistical resilience, and reduce overconcentration in key geographies such as Everett, Washington.

Objective and Rationale for Expansion

The purpose of this analysis is to identify an optimal U.S. location for a new Boeing production or integration facility. Expanding Boeing's domestic footprint serves several purposes: reducing geographic concentration risks, improving responsiveness to supply disruptions, and leveraging lower-cost environments. Additionally, expanding to a new location opens opportunities for tax advantages, improved workforce availability, and alignment with long-term strategic goals across Boeing's commercial and defense divisions.

Market Segment Analysis

Boeing operates in the aerospace and defense industry, competing primarily with companies such as Airbus, Lockheed Martin, Raytheon Technologies, and Northrop Grumman. These firms maintain extensive manufacturing networks and have increasingly diversified their global supply chains. While Airbus continues to lead in commercial aviation, Boeing's defense and space divisions are vital to its long-term viability.

Similarities:

- All are publicly traded, multinational companies.
- All operate in the aerospace and defense sector and have high R&D investments.
- All share exposure to government defense contracts and face supply chain complexity.

Differences:

- Boeing and Airbus are more heavily focused on commercial aviation, while Lockheed Martin and Northrop Grumman dominate defense technologies.
- Airbus has a stronger footprint in the EU, while Boeing remains U.S.-centric.

Facility Recommendation

A new facility is essential for:

- Reducing reliance on over-concentrated locations (e.g., Everett, WA)
- Enhancing geographic and logistical flexibility
- Tapping into regional labor and tax advantages
- Strengthening supply chain partnerships and workforce pipelines

In evaluating potential locations for the Boeing Company, I began with a broad scan of five cities: Everett (WA), Charleston (SC), Wichita (KS), Huntsville (AL), and Fort Worth (TX). These were assessed based on labor cost and availability, access to logistics and infrastructure, proximity to aerospace suppliers, risk diversification and alignment with Boeing's strategic goals. After a preliminary evaluation, **Charleston, SC** and **Wichita, KS** emerged as the top two contenders. Charleston offers an existing Boeing footprint and strong port infrastructure, while Wichita brings deep-rooted aviation expertise and cost advantages.

Criteria	Charleston, SC	Wichita, KS
Existing Boeing Presence	Yes	Indirect
Labor Cost	Moderate	Low
Tax Climate	Favorable	Strong
Logistics Access	Strong	Moderate
Aerospace Cluster	Yes	Yes
Risk of Overconcentration	Moderate	Low

Fig 1. Facility Location evaluation

Financial Analysis and Justification

The financial modeling for this analysis considered a 30-year facility life and cash flow analysis using CFAT. Supported by IRR, NPV, Annual Worth and Payback period. Revenue was projected at 0.9% of Boeing's annual revenue which is $693 million based on Boeing's historical margins and current strategic forecasts. These assumptions were guided by public financial statements (10-K), Boeing's operational benchmarks and data from the IRS and Bureau of Labor Statistics. All calculations were performed in a detailed Excel workbook included in the appendix. Using these assumptions, the results from Charleston and Wichita were as follows:

Metric	Charleston, SC	Wichita, KS
IRR (%)	4.48%	23.04%
NPV ($)	–$67.2 million	$185.0 million
Annual Worth ($)	–$7.13 million	$19.6 million
Payback Period (Years)	16 years	5 years

Fig 2. Financial Analysis Summary

The Wichita facility generates significantly better results across all financial measures. A 23.04% IRR reflects a strong return on capital, exceeding the 10% MARR comfortably. The payback period of just five years ensures faster liquidity and risk mitigation. In contrast, Charleston's extended payback period and negative annual worth make it less viable under the same revenue assumptions.

Wichita's competitive advantage lies in its deep integration within the aviation manufacturing industry. Known as the "Air Capital of the World," the city houses numerous tier-one and tier-two aerospace suppliers and maintains a skilled labor pool. Its workforce pipeline is fed by local institutions like Wichita State University and technical colleges that focus on aviation.

In financial terms, lower labor and capital costs, combined with a competitive tax environment, make Wichita a superior long-term choice. Boeing can also avoid over-concentrating operations in coastal areas, thereby improving operational resilience. The central U.S. location provides cost-effective logistics access to both coasts and key manufacturing hubs.

Charleston, while logistically strong and already home to Boeing's 787 operations, cannot deliver comparable financial performance in this scenario. Additionally, expanding to Wichita would geographically balance Boeing's production infrastructure and reduce dependency on a single coastal hub.

Conclusion

Based on the financial data, strategic alignment, and operational potential, Wichita, Kansas is recommended as the location for Boeing's new U.S. facility. The investment promises a strong IRR, rapid payback, and long-term positive cash flows. Boeing would benefit from cost savings, a mature aerospace ecosystem, and a more resilient production footprint. While Charleston remains a valued part of Boeing's existing operations, Wichita offers a higher return on investment and complements Boeing's national and international production strategy.

This report emphasizes the importance of balancing financial rigor with strategic foresight. Boeing's current challenges make capital discipline and operational flexibility more important than ever. Establishing a new facility in Wichita would help address both, accelerating recovery and enhancing future competitiveness. The strong financial metrics, along with Wichita's deep aviation heritage, make it the most compelling choice for expansion.

References

1. Boeing Annual Report Retrieved from https://www.boeing.com/investors/
2. IRS *MACRS Depreciation Tables (39-Year)*. Internal Revenue Service. https://www.irs.gov/publications/p946
3. U.S. Bureau of Labor Statistics. *Occupational Employment and Wage Statistics*. https://www.bls.gov/oes/
4. Quadrennial Supply Chain Review. (2021–2024). *U.S. Department of Commerce*

Appendices

Appendix A: Assumptions and its rationale

Assumptions	Charleston, SC	Wichita, KS
Estimated CAPEX ($) - Facility Construction and Setup Cost	165000000	160000000
Annual Revenue ($) (Projected Revenue = 0.9% of $77.8B)	693000000	693000000
Labor Cost per Employee	85000	80000
Labor Force	7000	7000
Total Labor Cost ($)	595000000	560000000
O&M Cost ($)	86625000	86625000
Tax Rate	26%	25%
Facility Size (sq ft)	1.5 million	1.5 million
Depreciation Method	MACRS 39-year	MACRS 39-year
Project Life	30 Years	30 Years
MARR	10%	10%

1. Revenue: **0.9%** of $77.8B = $693M/year
   - Realizing from project 1, Boeing has faced loss consecutively year after year since 2019. Before 2019 as well, since 2015, Boeing historically ran at around 5-10% net margin. Hence, even at peak performance years, new initiatives would realistically contribute <1% to company-wide revenue. To obtain realistic results considering Boeing's situation, I have assumed the closest lower value to 1%.
2. Facility Size: **1.5 million sq. ft**
   - The size is based on an average of Boeing's existing facilities. The Everett facility (4.3M sq ft) and South Carolina plant (1.2M sq ft) guided this decision, making 1.5M sq ft a scalable, mid-sized benchmark slightly leaning towards the smaller facility considering the company's liquidity.
3. Labor: 7,000 employees - Assumed based on Boeing's similar sized facilities labor force which is between 5000 to 10000 approximately.
4. Labor Cost per Employee: $85,000/yr at Charleston, SC and $80,000/yr at Wichita, KA.
   - Assumed based on industry standards and adjusted for geographic differences.
   - Labor cost is slightly lower in Wichita due to local cost-of-living and industry wage data
5. Estimated CAPEX ($): Facility Construction, Setup Cost and miscellaneous estimated at - $165M for Charleston, SC and -$160M for Wichita, KS.
6. Tax: 26% for South Carolina and 25% for Kansas considering the cumulative federal and state taxes respectively.
7. O&M: $86.625M/year based on aerospace industry benchmarks for similar scale.
8. Depreciation method (MACRS 39-years): Standard IRS method for non-residential real property, ensuring compliance and long-term write-off.

Appendix B: Financial Analysis

- CFAT calculations
- IRR, NPV, Payback Period
- Depreciation schedule (MACRS 39-Year)
- Annual Worth (AW) and cumulative cash flows

Financial Metrics Summary:

Metric	**Charleston, SC**	**Wichita, KS**
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Refer attached workbook for full financial analysis for projected results based on the assumptions considered.

Attachment: Project 2 Workbook_Boeing Company

Appendix C: Charts

Chart: Charleston, SC Vs Wichita, KS – CFAT Comparison